

120 Days of Quant Finance

Week 4 – Financial Theory

Day 2: Derivatives – Types and Basics

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January 2026

Introduction

Derivatives are financial instruments whose values are derived from underlying assets or market variables. Before addressing valuation and pricing, it is essential to understand the structural characteristics that define different types of derivative contracts.

This section presents a systematic classification of derivatives and introduces the core features that distinguish them.

Classification by Underlying Asset

Derivatives can be classified according to the nature of their underlying asset:

- **Equity derivatives:** stocks and equity indices.
- **Fixed-income derivatives:** interest rates and bonds.
- **Foreign exchange derivatives:** currency exchange rates.
- **Commodity derivatives:** physical commodities such as oil or metals.

Although the underlying assets differ, pricing principles remain unified across categories.

Classification by Contract Type

Forwards

A forward contract is an over-the-counter agreement to buy or sell an asset at a future time T for a predetermined price K . The payoff of a long forward at maturity is:

$$\text{Payoff} = S_T - K.$$

Futures

Futures contracts are standardized forward contracts traded on exchanges. They are subject to daily settlement and margin requirements, which introduce practical differences from forwards.

Options

Options provide the right, but not the obligation, to transact in the underlying asset at a specified price.

A European call option payoff is:

$$\max(S_T - K, 0),$$

while a European put option payoff is:

$$\max(K - S_T, 0).$$

Swaps

Swaps involve the exchange of cash-flow streams between two parties. Common examples include interest rate swaps, where fixed-rate payments are exchanged for floating-rate payments.

Linear and Nonlinear Derivatives

Derivatives may also be classified by the shape of their payoff functions:

- **Linear derivatives:** forwards, futures, and swaps.
- **Nonlinear derivatives:** options and structured products.

Nonlinear payoffs introduce convexity and complicate hedging and pricing.

Exercise Style

European Style

European derivatives can only be exercised at maturity, simplifying valuation.

American Style

American derivatives can be exercised at any time up to maturity, requiring more complex pricing techniques.

Contract Parameters

Derivative contracts are typically characterized by:

- Maturity T ,
- Strike price K ,
- Underlying price at maturity S_T .

A general derivative payoff can be expressed as:

$$\text{Payoff} = f(S_T; K, T).$$

Long and Short Positions

Every derivative contract has a long and a short position. The payoff of the short position is the negative of the long position payoff:

$$-f(S_T).$$

Pricing symmetry between long and short positions is enforced by no-arbitrage conditions.

Settlement Methods

Derivatives may be physically settled, involving delivery of the underlying asset, or cash settled, where only the net payoff is exchanged.

Conclusion

Understanding the types and structural features of derivatives is essential before developing pricing models. Clear specification of underlying assets, payoff functions, and contract parameters ensures consistency in valuation and risk management.

These foundations prepare the ground for the no-arbitrage principle and pricing methods introduced in subsequent sections.